

From Set-Level Inferability to Field-Level Risk: Budgeted Marginal Inference and an Amortized Attacker for Grading Power Market Data

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Abstract

Electricity-market data are graded and released field by field, yet inference risk arises from combinations: fields that reveal nothing alone can jointly expose unit outputs, line flows or load before release. We define field-level risk from set-level inferability, the out-of-sample accuracy that attackers re-trained on a set of fields reach for a sensitive target. The budgeted marginal risk $M^{(K)}$ of a field is its largest inferability gain over any background of at most K other fields, relative to data already public. It is a budget-constrained marginal contribution index, equals the minimal threshold-free safety margin, and flags exactly the members of minimal unsafe combinations of at most $K + 1$ fields; its witness background tells the data owner what not to co-release. To evaluate all small sets repeatedly, an amortized attacker—one network pre-trained under random visibility masks, queried through a closed-form readout per field set—scans the sets, and retraining a few backgrounds per field certifies the result. On four power data sets with rule-based field roles, including actual unit outputs from the Australian market, and 1.2×10^5 exhaustively retrained field sets as ground truth, single-field grading recovers 13–17% of the critical fields and leaves over 95% of dangerous combinations intact. Scan-certify recovers up to 88% of the critical fields without false positives, and withholding fields chosen from estimates alone breaks 94–99% of dangerous combinations with near-minimal withholding. Correlation, attribution and inference-graph scores are less effective at high thresholds; the per-set readout is the key component of the estimator; and $K = 2$ captures most of the marginal risk measured at $K = 3$, while disposal is best computed at $K = 3$.

Keywords: data classification and grading, inference attack, electricity

1. Introduction

Electricity markets publish a large and growing amount of operational data. Disclosure rules list, field by field, what must be released and when: forecasts and requirements before clearing, prices and their components right after clearing, and actual operating quantities on the following day [1]. Other quantities—the output of an individual generating unit, the hourly flow on a specific line, or the actual load before it is disclosed—are protected, either permanently or until a later release date. Data classification standards ask operators to grade each field and, explicitly, to consider how aggregation and processing change a field’s level [2]. How to do the latter is left open.

The difficulty is that inference risk is a property of field *combinations*. A price series reveals little about a unit’s output, and so does a renewable forecast; together they may reveal the residual demand that the unit is dispatched to meet. This is the classical inference and aggregation problem of database security [3, 4], and it is not hypothetical for power data: public prices alone can expose grid topology [5]. Grading fields one at a time is blind to it by construction. Recent work on data sensitivity assessment has recognized that a field’s risk depends on its inference relations to sensitive data and propagates risk along an inference graph [6]. What is still missing is a field-level risk that is *defined* by what an adversary can actually infer from a set of fields—so that it can be checked by an attack experiment rather than calibrated to expert scores—and that captures inference requiring several fields jointly.

This paper builds such a quantity and makes it computable at the scale of an audit. The auditor needs answers to three questions, and each has its own object:

1. *If a set S is published, how well can target y be inferred?* We use the out-of-sample R^2 of the strongest attacker in a fixed family retrained on S , made monotone by a closure that the adversary could itself implement. This set-level inferability $V_y(S)$ is the ground truth throughout.
2. *How risky is field i , and with which other fields?* We define the budgeted marginal risk $M_{i \rightarrow y}^{(K)}$, the largest increase of V_y that releasing i causes over any background of at most K other fields, relative to what

is already public. It is a budget-constrained marginal contribution index [7], i.e. a worst case rather than an average over contexts, in line with how quantitative information flow measures leakage [8, 9]. It comes with a witness background, an escalation over single-field risk, and an exact link to minimal unsafe field combinations.

3. *How can this be done for every field, every target and every change in what is public?* We pre-train one network under random visibility masks—an adversary that does not know in advance which fields it will obtain—and answer each query by a closed-form readout specific to the field set. The estimate is only used to scan; decisions are certified by retraining dedicated attackers on a handful of candidate backgrounds.

Contributions.

- *A checkable field-level risk.* We derive $M_{i \rightarrow y}^{(K)}$ from set-level inferability under a worst-case, bounded-background, public-baseline semantics, prove that it is the minimal threshold-free safety margin and that τ -criticality is equivalent to membership in a minimal unsafe set of size at most $K + 1$, and translate it into grading and disposal outputs (Sections 3–4).
- *An amortized attacker with scan-certify.* Masked pre-training plus a per-set closed-form readout estimates V_y for every small set in milliseconds; a common-mode error analysis explains why the field-level risk is estimated more accurately than the set values, and certification by retraining gives lower bounds that do not depend on the estimate (Section 5).
- *Evidence on four power data sets with rule-based field roles*—a synthetic grid with realistic time series (RTS-GMLC), the Australian National Electricity Market with actual unit outputs, and PJM and CAISO system data—with exhaustive retraining of a three-attacker family on all sets of up to four fields as ground truth. We compare against correlation-based, single-field, attribution-based and inference-graph scores, ablate every component of the estimator, and test robustness to attacker seeds, data splits, thresholds and the choice of K (Sections 6–10).

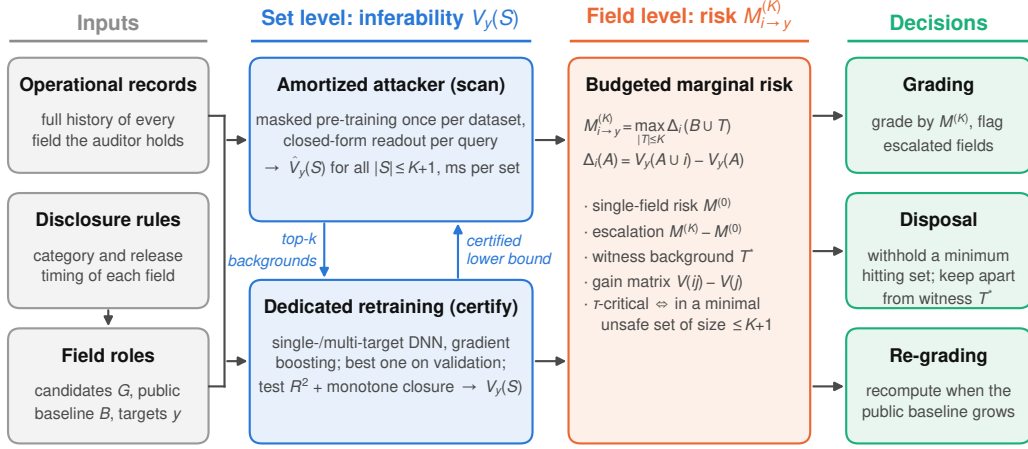


Figure 1: Audit framework. Disclosure rules fix field roles. Set-level inferability $V_y(S)$ is defined by dedicated retraining and scanned by an amortized attacker; the scan proposes, for every field, backgrounds that are certified by retraining. The field-level table drives grading, disposal and re-grading when the public baseline grows.

2. Related Work

Inference and aggregation in data release. That individually innocuous data items can jointly reveal protected information is an old observation in database security, where it is known as the inference and aggregation problem [3, 4]; the privacy literature supplies the best-known illustration, three demographic attributes that jointly single out most individuals [10]. In power systems, public market data have been shown to expose non-public physical information, for example grid topology from locational marginal prices [5]. Privacy mechanisms for metering data [11, 12] and formal models such as differential privacy [13] or k -anonymity [14] bound what can be learned about an individual record after a release mechanism is applied. They answer a different question from the one faced by a market operator, who must decide which aggregate operational *fields* may be published at all.

Data sensitivity assessment under inference leakage. Data classification and grading is performed field by field in practice, and the Chinese national standard explicitly asks that aggregation and processing effects be considered [2], without prescribing how. The closest work to ours is IGNN [6], which introduced data inference leakage into sensitivity assessment for cyber-physical power grids: fields are nodes of an inference graph whose edges encode physi-

cal or statistical inference relations, and sensitivity is propagated along these edges by a graph attention network from an intrinsic (expert-assigned) score. This established that the risk of a field depends on its inference relations to sensitive data. Two questions remain open in this line of work. First, the risk score is calibrated against expert labels, while its ultimate validation is an attack experiment in which a model is trained on a group of fields; the attack outcome itself is not the definition. Second, an edge connects two fields, whereas inference frequently requires several fields jointly, with none of them informative alone. We take the attack outcome on a field *set* as the definition of risk and derive the field-level quantity from it.

Feature importance. Turning set-level values into per-feature scores is the subject of feature attribution. Shapley-based measures [15, 16, 17, 18] average a feature’s marginal contribution over all contexts, which is the right answer for fair allocation but dilutes the credit of redundant features [19, 7]. Leave-one-covariate-out (LOCO) [20] and permutation importance [21, 22, 23] evaluate a single context, the full feature set, so a feature with a near-substitute receives almost no importance. The Marginal Contribution Index (MCI) [7] takes the *maximum* marginal contribution over contexts, is characterized axiomatically, and is immune to dilution. Security reasoning favours the worst case for the same reason quantitative information flow replaced Shannon entropy by min-entropy and gain functions [8, 9]: an average over contexts is not a guarantee. Our field-level risk is a budget-constrained MCI evaluated relative to already-public data.

Fast estimation of many refitted models. Computing the performance of a model refitted on many feature subsets is expensive, which has motivated lazy [24] and warm-started [25] estimators of variable importance, amortized explanation models [26, 27], and conditional generative models trained under random masks [28, 29, 30, 31]. Closed-form ridge heads on frozen features are the standard fast-adaptation device in meta-learning [32, 33], and the benefit of sharing a representation across tasks is well understood [34, 35, 36]. We combine these ingredients into an estimator of set-level inferability and, more importantly, use it inside a scan–certify loop so that decisions rest on retrained attackers rather than on the estimate.

3. Problem Formulation

3.1. Parties, field roles and threat model

The **auditor** is the data owner. It holds the complete history of all operational fields, including those it regards as sensitive, and grades fields offline before release. The **adversary** obtains published fields and is assumed to hold historical auxiliary labels of the sensitive target, from which it trains a supervised inference model; risk is therefore measured as out-of-sample *supervised* inferability. The adversary may combine the field under review with other fields it already holds.

Field roles are fixed by disclosure rules, not by data analysis. Let G be the candidate fields that the rules allow to be published, $B \subseteq G$ a *public baseline* that is already released ($B = \emptyset$ unless stated otherwise), $H = G \setminus B$ the fields under review, and Y_y a sensitive target. Section 6.1 instantiates these roles for four data sets from the categories and release timings of the Chinese electricity-market disclosure rules [1]. An ex-ante estimate of the *same physical quantity* as a target (e.g. a load forecast when the target is metered load) is removed from G , because it is a copy of the target rather than an inference channel.

3.2. Set-level inferability

Definition 1 (Population inferability).

$$V_y^*(S) = 1 - \inf_f \mathbb{E}[(Y_y - f(X_S))^2] / \text{Var}(Y_y),$$

the infimum taken over measurable f . It satisfies $V_y^*(\emptyset) = 0$, $0 \leq V_y^* \leq 1$ and is monotone in S .

Definition 2 (Empirical inferability). *The data are split into training and test rows, with a validation part inside training. For an attacker family $\mathcal{H}$, every $h \in \mathcal{H}$ is trained on X_S only; the attacker and its training epoch are selected on validation, and the test R^2 is reported: $V_y^{\text{ret}}(S) = R_{\text{test}}^2(\hat{h}_S)$ with $\hat{h}_S = \arg \max_{h \in \mathcal{H}, \text{epoch}} R_{\text{val}}^2(h; S)$.*

Finite samples, optimization and selection noise make V_y^{ret} slightly non-monotone. An adversary holding S can train on any subset of S and keep the best, so the operational object is the *monotone closure*

$$V_y(S) = V_y^{\text{ret}}(\hat{T}_S), \quad \hat{T}_S = \arg \max_{T \subseteq S} R_{\text{val}}^2(T), \quad (1)$$

Table 1: Requirements for a field-level inference-risk quantity. $\checkmark$: satisfied; $\circ$: partially; $-$: not satisfied.

Method	R1 checkable	R2 joint	R3 worst case	R4 no dilution	R5 bounded K	R6 baseline B
Correlation / mutual information	-	-	-	$\checkmark$	-	-
Single-field retraining $M^{(0)}$	$\checkmark$	-	-	$\checkmark$	-	$\circ$
Inference-graph propagation [6]	-	$\circ$	-	$\circ$	-	-
Shapley / SAGE [17]	$\checkmark$	$\checkmark$	-	-	-	-
LOCO / permutation	$\checkmark$	$\checkmark$	-	-	-	-
MCI [7]	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	-	-
$M_{i \rightarrow y}^{(K)}$ (this paper)	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$

where the subset is chosen on validation and the value is read on test. We write V_y for the closure throughout. It is the *ground truth* of this paper: every conclusion about risk is stated in terms of V_y computed by dedicated retraining.

3.3. Audit questions

The auditor asks three questions: (*Q1*) if a set S is published, how accurately can Y_y be inferred? (*Q2*) how risky is a single field i , why, and together with which other fields? (*Q3*) how can Q1–Q2 be answered for all fields and targets, repeatedly as the public baseline or target definitions change, without training a dedicated attacker for every set? Section 4 answers Q2 in terms of V_y , and Section 5 answers Q3.

4. Field-Level Inference Risk

4.1. Requirements

A field-level risk quantity derived from V_y should satisfy six requirements, listed in Table 1: (*R1*) it is checkable by an attack experiment; (*R2*) it sees joint inference, where fields are uninformative alone; (*R3*) it takes the worst case over what else the adversary holds; (*R4*) redundant substitutes do not dilute each other; (*R5*) the adversary’s background is bounded; (*R6*) risk is relative to data already public. Correlation and single-field retraining fail R2; propagation over pairwise inference graphs is not checkable in the sense of R1 and approximates R2 only through chains of pairs; Shapley-type averages fail R3–R4; LOCO and permutation importance evaluate a single context and fail R4. MCI satisfies R1–R4 but has no notion of R5–R6.

4.2. Definition

Definition 3 (Budgeted marginal inference risk). *For target y , public baseline B , fields under review $H = G \setminus B$ and budget K ,*

$$M_{i \rightarrow y}^{(K)} = \max_{T \subseteq H \setminus \{i\}, |T| \leq K} \Delta_i(B \cup T), \quad \Delta_i(A) = V_y(A \cup \{i\}) - V_y(A). \quad (2)$$

Any maximizer T^ is a witness background.*

$M_{i \rightarrow y}^{(0)} = V_y(B \cup \{i\}) - V_y(B)$ is single-field risk, the quantity current practice grades on; the *escalation* $M_{i \rightarrow y}^{(K)} - M_{i \rightarrow y}^{(0)} \geq 0$ is the risk that appears only in combination; the *gain matrix* $V_y(\{i, j\}) - V_y(\{j\})$ lists the increment of i over every single background.

Proposition 1 (Relation to MCI). *Let $v_B(T) := V_y(B \cup T) - V_y(B)$ for $T \subseteq H$. Then v_B is monotone, non-negative and $v_B(\emptyset) = 0$; $M_{i \rightarrow y}^{(K)} = \max_{|T| \leq K} [v_B(T \cup \{i\}) - v_B(T)]$ is the budget-constrained MCI of v_B ; for $K \geq |H| - 1$ and $B = \emptyset$ it is the MCI of V_y .*

Proof. Monotonicity of V_y (enforced by the closure) gives monotonicity of v_B , and $v_B(\emptyset) = 0$ by construction; the rest is substitution. $\square$

The axiomatic properties of MCI for monotone evaluation functions [7]—dummy, symmetry, the self-contribution lower bound $M_{i \rightarrow y}^{(K)} \geq M_{i \rightarrow y}^{(0)}$ and invariance to duplicated features—therefore transfer and are cited rather than claimed. The results below use the finite budget and are specific to the security reading.

4.3. Security meaning

Theorem 1 (Threshold-free safety margin). *Fix i, y, B, K . A number $m \geq 0$ satisfies “for every threshold τ and every background $|T| \leq K$: $V_y(B \cup T) \leq \tau - m \Rightarrow V_y(B \cup T \cup \{i\}) \leq \tau$ ” if and only if $m \geq M_{i \rightarrow y}^{(K)}$. For a fixed τ , the smallest m with this property can be strictly smaller than $M_{i \rightarrow y}^{(K)}$.*

Proof. If $m \geq M_{i \rightarrow y}^{(K)}$ then $V_y(B \cup T \cup \{i\}) = V_y(B \cup T) + \Delta_i(B \cup T) \leq (\tau - m) + m$. Conversely, for any T choose $\tau = V_y(B \cup T) + m$; the property gives $\Delta_i(B \cup T) \leq m$. Strictness: with $B = \emptyset, K = 1, V_y(\{i\}) = 0.1, V_y(\{j\}) = 0.5, V_y(\{i, j\}) = 0.9$ one has $M_{i \rightarrow y}^{(1)} = 0.4$, yet at $\tau = 0.95$ the necessary margin is 0. $\square$

Thus $M_{i \rightarrow y}^{(K)}$ is the answer to “by how much can publishing i raise the adversary’s accuracy, whatever it already holds within budget”—a property of the field that does not depend on the threshold later chosen for grading.

Definition 4 (Criticality and minimal unsafe sets). *Under baseline B , $E \subseteq H$ is a τ -minimal unsafe set (MUS) if $V_y(B \cup E) > \tau$ and $V_y(B \cup E') \leq \tau$ for every $E' \subsetneq E$. Field i is K -background τ -critical if some T with $|T| \leq K$, $i \notin T$, has $V_y(B \cup T) \leq \tau < V_y(B \cup T \cup \{i\})$.*

Theorem 2 (Criticality $\iff$ small MUS). *If V_y is monotone and $V_y(B) \leq \tau$, field i is K -background τ -critical if and only if it belongs to a MUS of size at most $K + 1$.*

Proof. ($\Leftarrow$) For a MUS $E \ni i$ with $|E| \leq K + 1$ take $T = E \setminus \{i\}$. ($\Rightarrow$) If $V_y(B \cup T) \leq \tau < V_y(B \cup T \cup \{i\})$, take an inclusion-minimal unsafe $E \subseteq T \cup \{i\}$; it is a MUS of size $\leq K + 1$. If $i \notin E$ then $E \subseteq T$ and monotonicity gives $V_y(B \cup T) \geq V_y(B \cup E) > \tau$, a contradiction. $\square$

Theorem 2 makes the field-level flag a faithful projection of the set-level danger structure: flagging the τ -critical fields misses no field that takes part in a dangerous combination of at most $K + 1$ fields, and the witness T^* names the combination. The same MUS list gives the smallest set of fields whose withholding breaks every dangerous small combination (a minimum hitting set), which we use for disposal. Monotonicity is essential: without the closure, a field can cross the threshold over a background that already contains an unsafe subset, and no MUS then contains it.

Proposition 2 (Closed form at $K = 1$). *With $I_{ij}^B = V_y(B \cup \{i, j\}) - V_y(B \cup \{i\}) - V_y(B \cup \{j\}) + V_y(B)$, $M_{i \rightarrow y}^{(1)} = M_{i \rightarrow y}^{(0)} + \max(0, \max_j I_{ij}^B)$.*

Hence a field escalates at $K = 1$ exactly when it is strictly super-additive with some partner.

4.4. Choice of K

$M_{i \rightarrow y}^{(K)}$ is non-decreasing in K , so K trades coverage against the cost of evaluating all sets of size $\leq K + 1$. We use $K = 2$, for three reasons. The threat model is an adversary holding a handful of other published series, not the whole catalogue. Empirically, high-order interactions are rare in engineered systems (the hierarchy principle of factorial experiments [37]), and models restricted to pairwise interactions are close to unrestricted ones

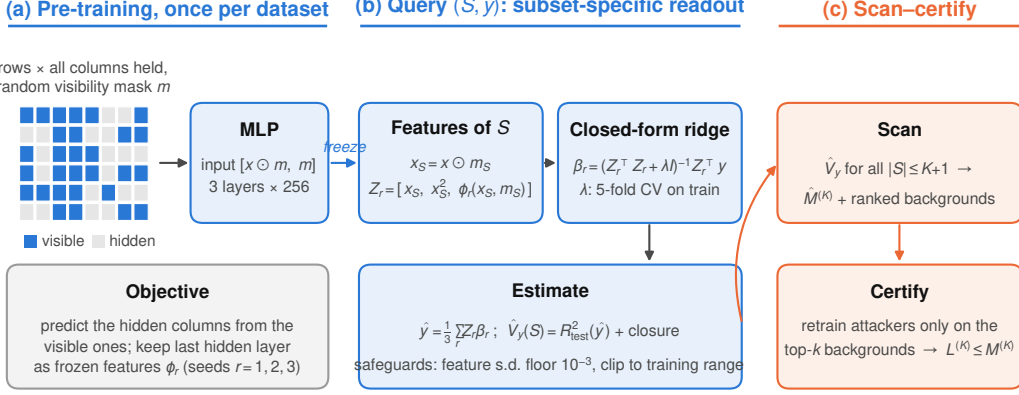


Figure 2: The amortized attacker. (a) One backbone is pre-trained under random visibility masks on the auditor’s data and frozen. (b) A query (S, y) solves a ridge readout on the features of S in closed form, per seed, and averages the predictions. (c) The scan proposes backgrounds per field; retraining only these certifies a lower bound on $M_{i \rightarrow y}^{(K)}$.

on tabular data [38]. Finally, we measure saturation directly on every data set (Section 10). Two caveats apply. There is no general theorem—a parity function of $K + 2$ fields defeats any fixed K —and saturation of the marginal gain does not imply that no larger set crosses the threshold. Section 10 shows that both matter in practice: $K = 2$ captures most of the marginal risk, but disposal decisions, which must break every dangerous combination, are better computed at $K = 3$.

5. An Amortized Attacker for Set-Level Inferability

5.1. Why an estimator is needed

Evaluating $M_{i \rightarrow y}^{(K)}$ for all fields at $K = 2$ requires V_y on every set of size at most three; with 35 candidates this is 7,175 sets per target, each requiring a family of attackers to be trained. The table has to be recomputed whenever the public baseline grows, a target is redefined, or a what-if release is examined. We therefore build one model that answers $V_y(S)$ for any S quickly (Fig. 2), and we make sure that no conclusion depends on its accuracy alone (Section 5.6).

5.2. Masked pre-training

The adversary does not know in advance which fields it will obtain. We model this by pre-training one network under random visibility masks. A

mask $m \in \{0, 1\}^p$ is drawn by first sampling the number of visible fields uniformly and then a uniform subset of that size; the network $f_\theta(x \odot m, m)$ receives the masked values and the mask itself, so that a hidden field is distinguishable from a zero value. Two supervision choices are used. *Target supervision* predicts the designated sensitive targets from the visible candidates. *Masked reconstruction* predicts every hidden column of the auditor’s data from the visible ones [30, 31]; it involves no designated target, so one pre-training serves any later definition of targets and candidates. The last hidden layer (256 units) is kept as a frozen feature map $\phi_\theta(x \odot m, m)$. Early stopping uses the validation rows only.

5.3. Subset-specific closed-form readout

For a query (S, y) with mask m_S and seed r , the feature block is

$$Z_r(S) = [x \odot m_S, (x \odot m_S)^2, \phi_{\theta_r}(x \odot m_S, m_S)], \quad (3)$$

standardized with training statistics. A ridge readout

$$\beta_r(S) = (Z_r^\top Z_r + \lambda I)^{-1} Z_r^\top y \quad (4)$$

is solved in closed form on the training rows, with λ chosen by five-fold cross-validation inside the training rows. The predictions of $R = 3$ independently pre-trained backbones are averaged, $\hat{y} = R^{-1} \sum_r Z_r^{\text{test}} \beta_r(S)$, and $\hat{V}_y(S) = R_{\text{test}}^2(\hat{y})$, followed by the monotone closure. Two numerical safeguards matter on real data: a floor of 10^{-3} on feature standard deviations, and clipping test features to their training range. Without them, units that are almost constant on the training rows (inactive ReLUs) are amplified on test rows, a few predictions explode and $\hat{V}$ collapses to zero for those sets; since $M_{i \rightarrow y}^{(K)}$ is a maximum over differences, such collapses turn directly into inflated risk (Section 9). No test label enters pre-training, the readout, the choice of λ or the closure.

5.4. Why the readout is solved per subset

A cheaper alternative lets the network output the prediction directly, $\hat{y} = f_\theta(x \odot m_S, m_S)$, and reads $\hat{V}$ off it. Then one set of output weights serves every conditioning pattern, i.e. the attacker is forced to combine every subset of fields in the same way—the very effect that combination risk is about. Solving the readout per subset keeps the expensive representation shared and the combination rule specific.

5.5. Error propagation

Write the estimate as $\hat{V}(S) = V(S) + b + \varepsilon(S)$, where b is a bias common to the sets of interest and $\varepsilon(S)$ the set-specific part. Since a marginal is a difference of two nested sets, $\hat{\Delta}_i(T) - \Delta_i(T) = \varepsilon(T \cup \{i\}) - \varepsilon(T)$ and

$$|\hat{M}_{i \rightarrow y}^{(K)} - M_{i \rightarrow y}^{(K)}| \leq \max_{|T| \leq K} |\varepsilon(T \cup \{i\}) - \varepsilon(T)|, \quad (5)$$

independently of b . A readout on a frozen representation typically underestimates the fully trained attacker by a similar amount on nested sets, which is exactly a common-mode b ; this is why an estimator can be biased in V yet accurate in M . The quantity to control is the *marginal* error, which we report throughout.

5.6. Scan and certify

The estimator is used to *scan*: it produces $\hat{M}^{(K)}$ for every field together with a ranked list of candidate backgrounds. Decisions are then *certified*: for each field, dedicated attackers are retrained on the top- k backgrounds only, and $L_{i \rightarrow y}^{(K)} = \max_{T \in \text{top-}k} \Delta_i(B \cup T) \leq M_{i \rightarrow y}^{(K)}$ is a lower bound computed entirely from retrained attackers. The ratio $\sum_i L / \sum_i M$ measures how much of the true risk the scan recovers at a certification cost of k retrained backgrounds per field instead of all $\binom{|H|-1}{\leq K}$.

6. Experimental Setup

6.1. Data sets and rule-based field roles

Table 2 summarizes the four data sets. In all of them, candidate fields are information that the disclosure rules [1] release *before* the target, and targets are either specific information (unit output, Art. 17(3)), quantities not on the disclosure list at hourly resolution (hourly line flow; only daily averages of important lines are listed), or actual operating quantities released on the following day (actual load, total generation, tie-line exchange; Appendix items 6.57–6.66). Using the release timing to assign roles does not involve any temporal split of the data; all splits are random.

RTS-GMLC. The IEEE Reliability Test System 2019 update [39] provides a 73-bus network with one year of realistic hourly load, wind, solar and hydro series. We clear an hourly DC optimal power flow with piecewise-linear offers built from the published heat-rate curves (8,784 hours, 26.7%

Table 2: Data sets. n : train/test rows; p : candidate fields; sets: field sets with exact three-attacker ground truth.

Data set	Targets (rule basis)	n	p	sets	max size
RTS-GMLC	unit 223, unit 221 (specific info.); line C35 (not listed)	6149/2635	23	10,902	4
NEM 2024	units BW01, TUMUT3, PPCCGT (specific info.)	6149/2635	30	31,930	4
PJM (load)	metered load (next-day)	4589/1967	22	9,108	4
PJM (gen./interch.)	total generation, net interchange (next-day)	4589/1967	24	12,950	4
CAISO	actual load (next-day)	6105/2617	35	59,535	4

congested). Candidates are the 23 fields a market would publish early (zonal load forecasts, renewable and hydro forecasts, reserve requirements, system energy price and nine nodal prices, three congestion indicators); targets are the outputs of a coal and a combined-cycle unit and the hourly flow on internal line C35.

NEM. One year (2024) of the Australian National Electricity Market [40], averaged to hours. Candidates are regional demand, available capacity, semi-scheduled wind and solar forecasts, regional prices and interconnector flows (30 fields); targets are the actual outputs of a coal unit (BW01), a hydro unit (TUMUT3) and a gas combined-cycle unit (PPCCGT). Unit output is public in the NEM but is specific information under the Chinese rules; this data set therefore contains a genuinely sensitive field observed in the real world.

PJM and CAISO. System-level hourly data for one year. Targets are the actual quantities released on the next day: PJM metered load, total generation and net actual interchange, and CAISO actual load. Candidates are forecasts, day-ahead and real-time prices and their components, ancillary-service requirements and cleared volumes, and schedules (22–35 fields). Fields released on the same day as the target (fuel-mix generation, actual renewable output, other balancing areas’ actual load) are excluded, as are the load forecasts of the target area, which are ex-ante copies of the target (Section 3). Near-copies that are distinct quantities (e.g. sub-area forecasts for CAISO load) are kept; removing them is analysed as a sensitivity case.

6.2. Ground truth

All variables are standardized with training statistics; 70/30 random train/test splits are used, with 15% of the training rows as validation. The attacker family has three members: a single-target multilayer perceptron (two hidden layers of 128 units, dropout 0.15, Adam, early stopping on validation), a multi-target perceptron trained on all targets of a data set, and

histogram gradient boosting with two configurations. For every set of up to four fields, every member is trained on that set alone; the member and epoch are chosen on validation and the test R^2 is reported, followed by the monotone closure (1). This amounts to 124,425 field sets and several hundred thousand trained models. Neither the estimator nor any baseline has access to these values; they are used only for evaluation.

6.3. Baselines

Field scores compared with $M^{(2)}$ are: Pearson r^2 , squared Spearman correlation, mutual information [41], distance correlation [42], single-field re-training $M^{(0)}$ (current practice), LOCO [20], permutation importance [21], SAGE with marginal imputation [17], and an inference-graph score that follows the propagation idea of [6] without its expert inputs: edge weights are the single-field predictability of one field from another (gradient boosting, validation R^2), the target node carries risk one, and risk is propagated over three hops with decay 0.5. All baseline scores are computed on training rows only.

Estimators compared with ours are: linear and quadratic readouts without a backbone, an in-set quadratic dictionary (values, squares and pairwise products of the fields in S), the shared output head of the same backbone, an untrained (randomly initialized) backbone with the same readout, and warm-start fine-tuning of the whole network for each set [25].

6.4. Metrics

For set values: mean absolute error of $\hat{V}$. For field risk: mean absolute error and bias of $\hat{M}^{(2)}$, Spearman correlation across fields, and the certified ratio $\sum_i L_{i \rightarrow y}^{(2)} / \sum_i M_{i \rightarrow y}^{(2)}$ when k backgrounds per field are retrained. For decisions at thresholds $\tau \in \{0.5, 0.7\}$: recall and precision of critical fields; the number of fields withheld and the number of minimal unsafe sets of size ≤ 3 left unbroken.

7. Results I: Combination Risk that Single-Field Grading Misses

Table 3 and Fig. 3 report, for each of the ten targets, how many candidate fields are unsafe on their own and how many become critical only in combination. Three patterns stand out.

Most critical fields are invisible to single-field grading. Of the 264 field-target pairs, 256 are critical at $\tau = 0.5$ with at most two background fields,

Table 3: Combination risk that single-field grading misses. V_1 : best single field; $V_{\leq 3}$: best set of at most three fields; alone: fields with $V(\{i\}) > \tau$; comb.: fields safe alone but τ -critical with at most two background fields; MUS: minimal unsafe sets of size ≤ 3 ; left: MUS not broken after withholding every field that is unsafe alone; hit: minimum number of fields whose withholding breaks every MUS.

Target	p	V_1	$V_{\leq 3}$	$\tau = 0.5$					$\tau = 0.7$				
				alone	comb.	MUS	left	hit	alone	comb.	MUS	left	hit
RTS unit 223	23	1.00	1.00	10	13	60	50	15	10	6	16	6	11
RTS unit 221	23	0.99	0.99	10	13	52	42	15	10	9	34	24	13
RTS line C35	23	0.52	0.86	4	19	123	119	13	0	21	82	82	7
NEM BW01	30	0.49	0.65	0	30	266	266	4	0	0	0	0	0
NEM TUMUT3	30	0.38	0.53	0	22	23	23	1	0	0	0	0	0
NEM PPCCGT	30	0.55	0.80	2	28	215	213	6	0	10	8	8	1
PJM load	22	0.48	0.80	0	22	237	237	9	0	14	44	44	3
PJM total gen.	24	0.97	0.99	2	22	179	177	11	2	18	39	37	5
PJM net interch.	24	0.41	0.71	0	24	102	102	3	0	3	1	1	1
CAISO load	35	0.82	0.93	5	30	542	537	17	3	32	335	332	10

and 223 of these are safe on their own; at $\tau = 0.7$ the figures are 138 and 113. For four targets (NEM BW01 and TUMUT3, PJM load and PJM net interchange) no candidate exceeds $\tau = 0.5$ alone, yet 22 to 30 candidates take part in a dangerous combination; at $\tau = 0.7$ the same holds for RTS line C35, where 21 of 23 fields are critical and none is unsafe alone. Withholding every field that is unsafe alone breaks almost none of the minimal unsafe sets (column “left”): 1,766 of 1,799 at $\tau = 0.5$ and 534 of 559 at $\tau = 0.7$ remain.

Where single fields do leak, combinations still dominate. Some targets are exposed by a single close proxy: RTS unit outputs are almost determined by a nodal price ($V_1 = 0.99$), and PJM total generation by the system load forecast (0.97). For these, $M^{(0)}$ already flags the proxy, and the remaining risk is again combinatorial (e.g. 22 of 24 PJM fields are combination-critical). Removing such near-proxies from the candidate pool (Section 6.1) lowers the best single field to 0.42 for both PJM total generation and CAISO load but leaves every remaining field combination-critical at $\tau = 0.5$ (22/22 and 26/26), so the conclusion does not depend on where the proxy boundary is drawn.

The witnesses are physically meaningful. Fig. 4 shows the field profile of RTS line C35 and the gain matrix of NEM unit PPCCGT. The largest escalations come from combinations that reconstruct a residual quantity: the SP15 day-ahead energy price rises from $M^{(0)} = 0.14$ to $M^{(2)} = 0.51$ for CAISO load with the NP15 wind and SP15 solar forecasts as witnesses (price and renewables determine net load); the South Australian wind forecast rises

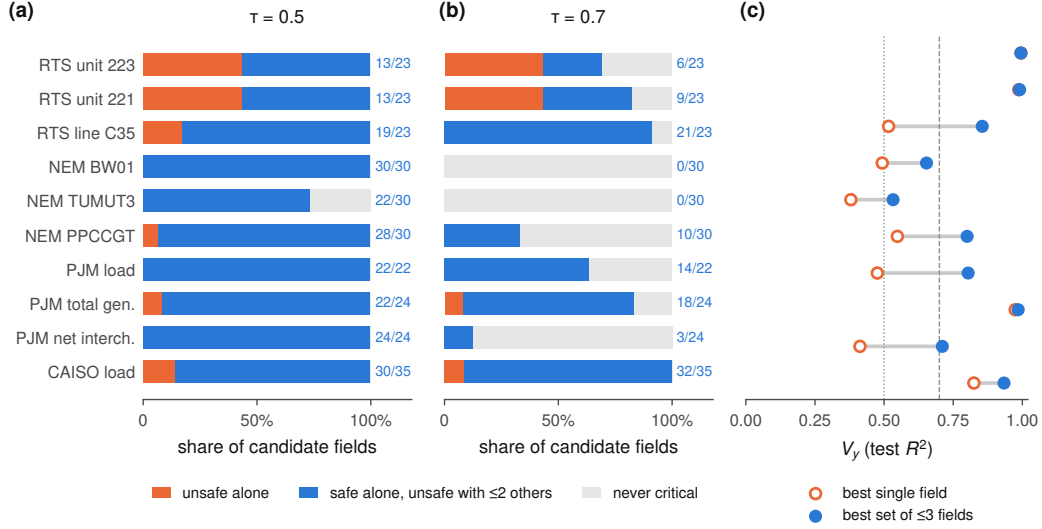


Figure 3: Share of candidate fields that are unsafe alone (orange), safe alone but critical with at most two background fields (blue), or never critical (grey), at (a) $\tau = 0.5$ and (b) $\tau = 0.7$; numbers give combination-critical fields over all candidates. (c) Best single field versus best set of at most three fields.

from 0.19 to 0.60 for PPCCGT together with the regional available capacity and solar forecast (the residual demand the gas unit is dispatched for); the price at RTS bus 303 rises from 0.36 to 0.65 for the C35 flow together with a zonal load forecast and the congestion indicator of the neighbouring line C6. The gain matrix also shows that the same field is harmless next to some partners and decisive next to others, which is what a grade that ignores the background cannot represent.

8. Results II: Comparison of Field-Level Risk Definitions

Mechanism. Fig. 5 uses synthetic data with a known structure, $Y = aAB + bP + \sigma\epsilon$, where A and B act only jointly, C_1 and C_2 are two noisy copies of P , D is a weak copy of P and N is noise; all 63 sets are retrained. Correlation, single-field retraining and pairwise inference-graph propagation give A and B a score of zero although $\{A, B\}$ reaches 0.54. LOCO gives C_1 and C_2 almost zero because each substitutes for the other, and LOCO, permutation importance and SAGE all give D zero because the full model relies on C_1, C_2 instead—yet $\{A, B, D\}$ crosses $\tau = 0.7$ while $\{A, B\}$ does not, so D is critical. SAGE splits credit within both pairs. $M^{(2)}$ assigns A, B, C_1, C_2 their full

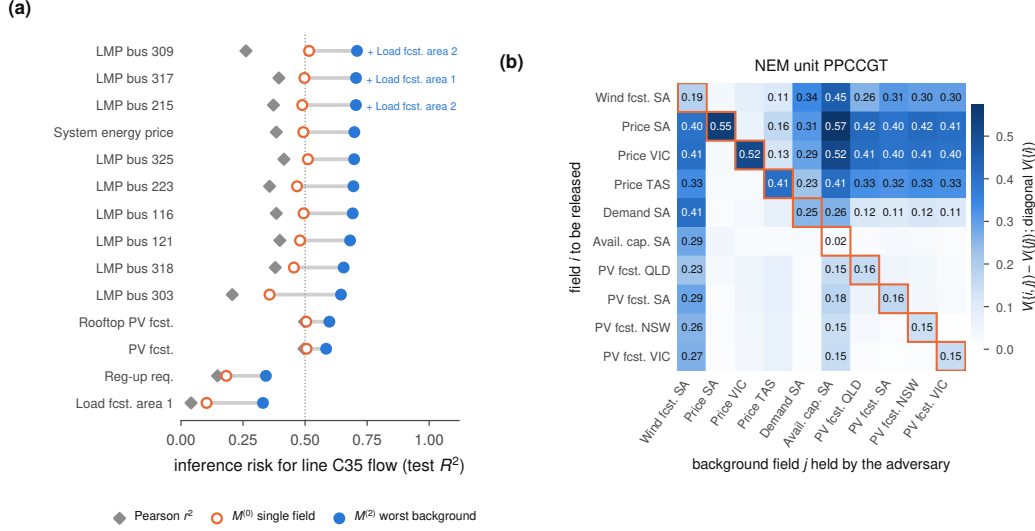


Figure 4: (a) Field profile for the hourly flow on RTS line C35 (14 fields with the largest $M^{(2)}$): Pearson r^2 , single-field risk and worst-background risk, with the witness background for the three riskiest fields. (b) Gain matrix for NEM unit PPCCGT: entry (i, j) is $V(\{i, j\}) - V(\{j\})$, the diagonal is $V(\{i\})$; values below 0.10 are not printed.

worst-case contribution and gives D a non-zero risk with the witness $\{A, B\}$. This is the behaviour predicted by Table 1: averages dilute, single contexts miss substitutes, and pairwise edges miss joint inference.

Decisions. The deliverable of grading is a decision, so we first compare flagging rules (Table 4, Fig. 6a). The single-field rule recovers 13% of the critical fields at $\tau = 0.5$ and 17% at $\tau = 0.7$. The scan on amortized estimates recovers 93% and 83%; with certification, which only accepts a field after dedicated attackers retrained on one of its three best-scanned backgrounds confirm the threshold crossing, precision is exactly one by construction and recall is 88% ($\tau = 0.5$, $\delta = 0$) and 80% ($\tau = 0.7$, $\delta = 0.05$). The margin δ compensates for the common-mode underestimation of the estimator (Section 9), which matters near high thresholds.

For disposal, the relevant question is how many fields must be withheld to break every dangerous combination. Withholding what single-field grading flags leaves 98% ($\tau = 0.5$) and 96% ($\tau = 0.7$) of the minimal unsafe sets intact. Two strategies that select fields from amortized estimates only—a minimum hitting set of the estimated minimal unsafe sets, and a greedy rule that withholds the field with the largest $\hat{M}^{(2)}$ and recomputes $\hat{M}^{(2)}$ on the

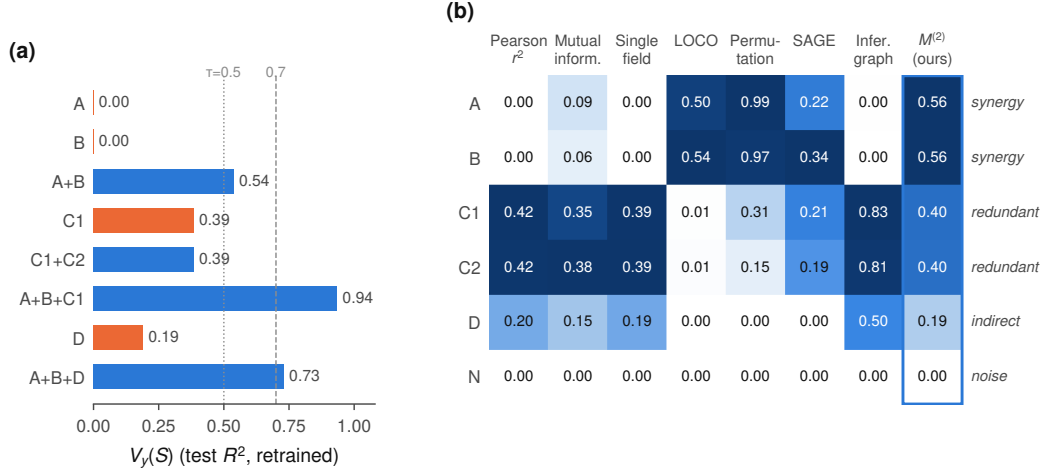


Figure 5: Synthetic mechanism example. (a) Retrained inferability of selected sets. (b) Scores of eight methods (colour normalized per column); the last column is $M^{(2)}$.

Table 4: Decisions over the ten targets. Top: recall and precision of τ -critical fields. Bottom: fields withheld (optimum in parentheses) and share of true minimal unsafe sets left unbroken. Selection uses estimates only; retrained values are used for certification and evaluation. δ : conservative margin of the scan.

Flagging rule	$\tau = 0.5$		$\tau = 0.7$	
	recall	precision	recall	precision
Single-field rule $V(\{i\}) > \tau$	0.13	1.00	0.17	1.00
Scan on $\hat{V}$, $\delta = 0.05$	0.93	0.99	0.83	0.80
Scan + certify, $\delta = 0$	0.88	1.00	0.75	1.00
Scan + certify, $\delta = 0.05$	0.77	1.00	0.80	1.00
Withholding strategy	withheld	left	withheld	left
Single-field grading	3.3 (9.4)	98.2%	2.5 (5.1)	95.5%
Hitting set of MUS on $\hat{V}$, $\delta = 0.05$	9.4 (9.4)	5.8%	5.3 (5.1)	1.8%
Adaptive $\hat{M}^{(2)}$ greedy, $\delta = 0.05$	10.6 (9.4)	2.3%	6.1 (5.1)	1.3%

remaining fields—leave 1.3–5.8% intact while withholding at most 1.2 fields more than the optimum computed from the ground truth (Fig. 6c). The greedy rule exploits the relative nature of $M_{i \rightarrow y}^{(K)}$: once a field is withheld, the risk of the others changes, and $M_{i \rightarrow y}^{(K)}$ is recomputed with respect to what remains.

Table 5: Ranking quality over the ten targets. R-precision: share of true critical fields among the top- c fields, c being the true number of critical fields (a threshold that favours the baselines). Excess: fields withheld in score order until no MUS remains, minus the optimum.

Score	R-precision		excess withheld	
	$\tau = 0.5$	$\tau = 0.7$	$\tau = 0.5$	$\tau = 0.7$
Pearson r^2	0.977	0.815	5.4	3.0
Spearman ρ^2	0.977	0.767	3.5	2.8
Mutual information	0.977	0.866	2.0	2.1
Distance correlation	0.982	0.833	3.0	2.0
Single-field $V(\{i\})$	0.982	0.803	1.6	1.9
LOCO	0.977	0.727	11.0	8.0
Permutation importance	0.968	0.795	8.9	4.7
SAGE	0.968	0.769	8.6	6.3
Inference-graph propagation	0.982	0.830	5.6	2.7
$\hat{M}^{(2)}$ (amortized, no retraining)	0.982	0.903	3.0	0.9
$M^{(2)}$ scan-certify ($k = 3$)	0.982	0.888	1.6	0.4
$M^{(2)}$ exact	0.977	0.894	1.5	0.4

Ranking. Table 5 compares the scores as rankings. At $\tau = 0.7$, $\hat{M}^{(2)}$ has the highest R-precision (0.903, against 0.866 for mutual information, 0.830 for the inference graph, 0.803 for single-field retraining, 0.769 for SAGE and 0.727 for LOCO), and withholding fields in the order of exact or certified $M^{(2)}$ exceeds the optimum by only 0.4 fields. At $\tau = 0.5$ almost every field is critical and every score reaches an R-precision of about 0.98; here single-field retraining ranks as well as $M^{(2)}$. We report this plainly: as a *ranking*, single-field risk is a strong baseline when most fields are dangerous; the advantage of $M_{i \rightarrow y}^{(K)}$ lies in decisions, disposal and high thresholds. Attribution scores computed on one fitted model (LOCO, permutation importance, SAGE) are the weakest throughout, because they measure what that model uses rather than what an adversary can infer.

9. Results III: The Amortized Attacker

Fidelity. Fig. 7 compares the amortized estimates with retrained ground truth for all sets of size ≤ 3 . The estimate of V is biased downward, visibly so for PJM, where the training set is smallest; averaged over the ten

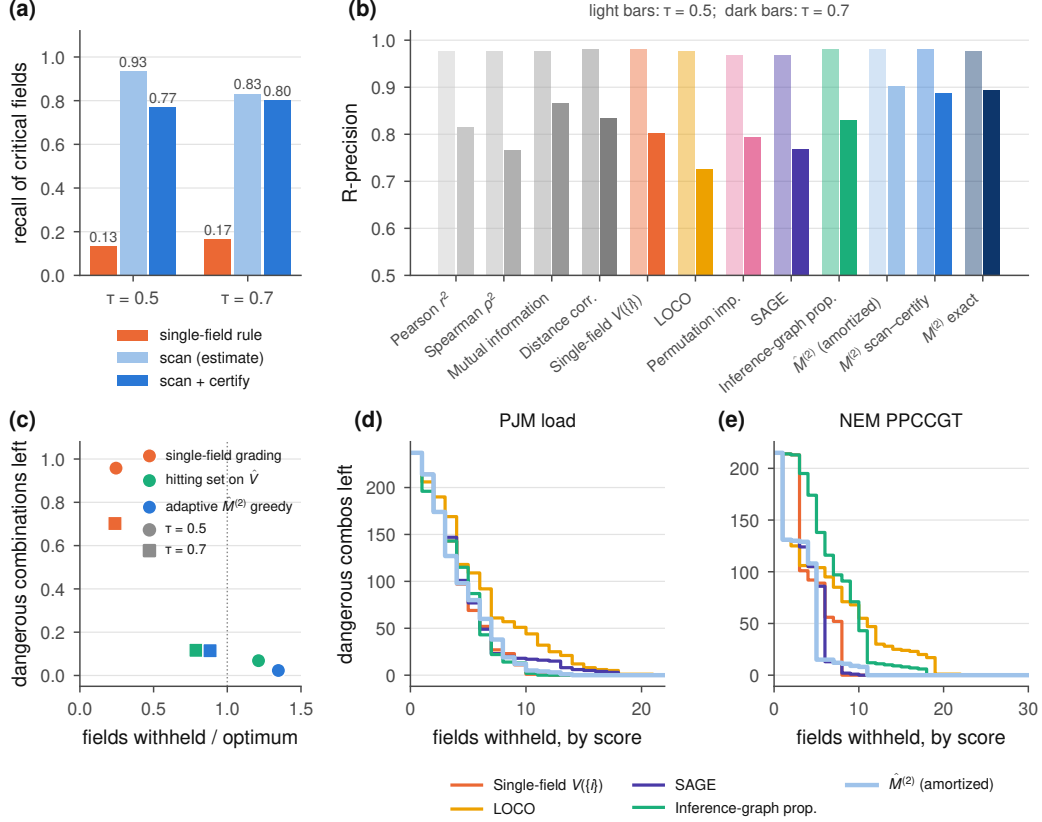


Figure 6: Definition comparison over ten targets. (a) Recall of critical fields by the single-field rule, the scan and scan + certify ($\delta = 0.05$). (b) R-precision of twelve scores (light: $\tau = 0.5$, dark: $\tau = 0.7$). (c) Fields withheld relative to the optimum versus share of minimal unsafe sets left, for single-field grading and two estimate-driven strategies. (d, e) Minimal unsafe sets left when fields are withheld in score order, for two targets.

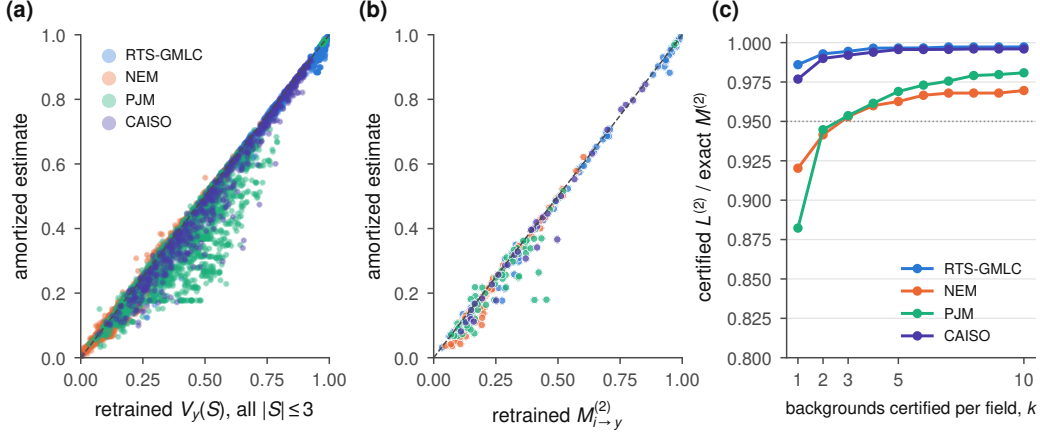


Figure 7: Amortized estimates against retrained ground truth. (a) $\hat{V}_y(S)$ for sets of size ≤ 3 (random subsample). (b) $\hat{M}^{(2)}$ for every field and target. (c) Certified lower bound over exact $M^{(2)}$ when the top- k scanned backgrounds per field are retrained.

targets the error is 0.026 with a signed bias of -0.024 , i.e. almost entirely common-mode. As predicted by (5), the field-level risk is estimated more accurately than the set values it is built from: the mean absolute error of $\hat{M}^{(2)}$ is 0.021 and its Spearman correlation with the exact $M^{(2)}$ across fields is 0.957 (Table 6). Certification recovers most of the exact risk from very few retrained backgrounds: retraining the single best-scanned background per field yields a lower bound equal to 93.4% of the exact $M^{(2)}$ on average, three backgrounds yield 97.0%, and on RTS-GMLC and CAISO the top-3 ratio exceeds 99% (Fig. 7c). For a field set of 35 candidates this replaces the $1 + 34 + \binom{34}{2} = 596$ backgrounds per field of an exhaustive evaluation by three.

What matters in the estimator. Table 6 and Fig. 8(a–c) separate the components. The largest single effect is solving the readout per field set: with the same backbones, reading the prediction off a shared output head gives an $M^{(2)}$ error of 0.057, against 0.021 for the per-set readout, and a lower certified ratio. The second is a learned nonlinear representation: without a backbone, a linear readout (0.144) or an in-set quadratic dictionary (0.077) is far less accurate. Masking during pre-training is necessary—a backbone trained on fully visible inputs is worse than the per-set readout on any other backbone (0.036, Spearman 0.928)—but the masking distribution is not critical (Bernoulli versus uniform visible count: 0.024 versus 0.023). The numeri-

Table 6: Estimator comparison and ablation, averaged over the ten targets and all sets of size ≤ 3 . Ablations use one seed and should be compared with “ours, one seed”. Cert.@ k : certified $L^{(2)}$ over exact $M^{(2)}$ when k backgrounds per field are retrained.

Estimator	MAE V	MAE Δ	MAE $M^{(2)}$	bias $M^{(2)}$	Spearman	cert.@1	cert.@3
Linear readout on $x \odot m$ (no backbone)	0.212	0.068	0.144	-0.139	0.737	0.803	0.878
Readout on $[x, x^2]$ (no backbone)	0.132	0.049	0.096	-0.089	0.854	0.859	0.912
In-set quadratic dictionary (no backbone)	0.120	0.045	0.077	-0.066	0.870	0.872	0.932
Shared output head (3 seeds)	0.067	0.030	0.057	-0.054	0.930	0.883	0.939
Untrained random backbone + readout	0.027	0.017	0.022	-0.015	0.959	0.906	0.971
Pre-training without masks	0.032	0.020	0.036	-0.004	0.928	0.898	0.963
Bernoulli(0.5) masks	0.030	0.019	0.024	-0.018	0.958	0.932	0.969
Readout on ϕ only	0.030	0.019	0.025	-0.016	0.950	0.927	0.964
No numerical safeguards	0.031	0.020	0.036	+0.009	0.904	0.825	0.946
Ours , one seed	0.030	0.019	0.023	-0.019	0.953	0.936	0.967
Ours , three seeds (main)	0.026	0.017	0.021	-0.018	0.957	0.934	0.970

cal safeguards remove the overestimation of $M^{(2)}$ on real data (bias +0.009 without them, certified top-1 ratio 0.825 instead of 0.936). Averaging three seeds reduces the V error from 0.030 to 0.026.

One result qualifies the role of pre-training and we state it explicitly. A randomly initialized, untrained backbone with the same per-set readout estimates V and $M^{(2)}$ about as well as the pre-trained one (0.027 and 0.022). The per-set readout on random ReLU features of the masked input is itself a random-feature regression [43], which is already a good model of a set’s inferability. What pre-training adds is the ordering of the largest marginals, which is what certification relies on: the certified top-1 ratio is 0.936 for the pre-trained and 0.906 for the random backbone, and 0.940 versus 0.694 on PJM load, the hardest target. We therefore attribute the accuracy of the estimator primarily to the per-set readout and the conditional feature map, and the reliability of the scan to masked pre-training.

Warm-start fine-tuning of the whole network for each set, the natural fast alternative to retraining [25], is less accurate than the closed-form readout on nine of ten targets (on 150 random sets per data set; e.g. 0.028–0.044 versus 0.013–0.021 on RTS-GMLC) while being slower.

One pre-training for all target definitions. In practice the auditor’s target list and candidate pool change—a field is reclassified, a new target is added. With masked reconstruction over all 53 columns of a market, one backbone serves every such definition through the readout alone. Table 7 compares it with backbones supervised on each group’s own targets. The all-column backbone is more accurate for V (0.040 versus 0.049) and slightly more accurate for

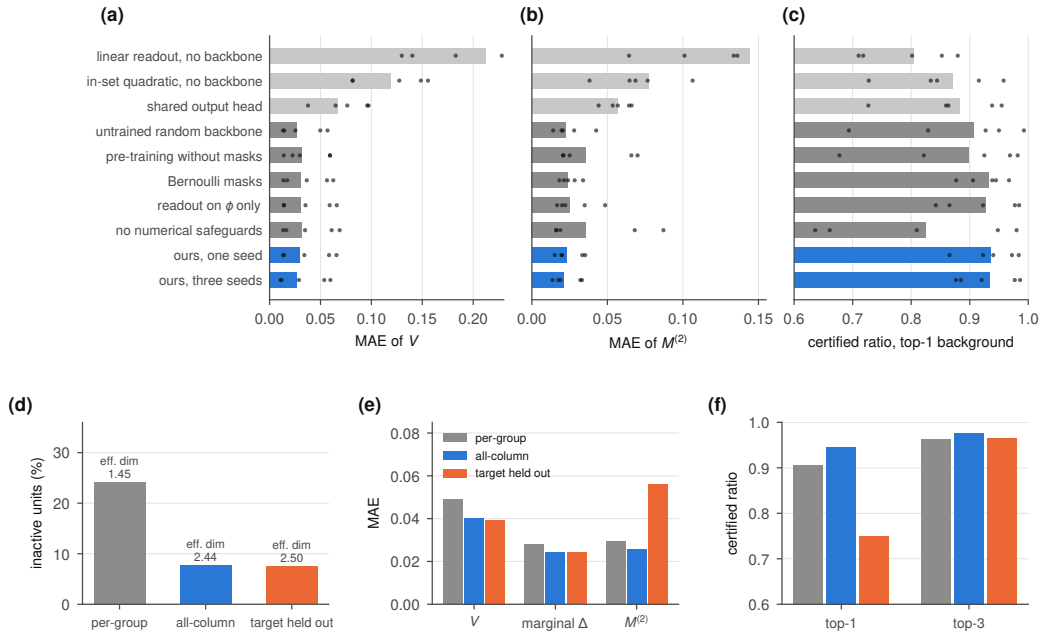


Figure 8: (a–c) Estimator comparison and ablation over ten targets (bars: mean; dots: data sets); light grey: no per-set readout or no backbone; dark grey: one-seed ablations; blue: ours. (d–f) Backbone scope on the four PJM/CAISO targets: inactive hidden units and effective dimension, error of V , of marginals and of $M^{(2)}$, and certified ratios.

Table 7: Backbone scope on the four PJM/CAISO targets (same readout). Inactive: hidden units with zero variance on the training rows; eff. dim: participation ratio of the feature covariance; common-mode: $|\bar{e}|/|e|$ of the V error e .

Backbone	inactive	eff. dim	MAE V	common-mode	MAE Δ	MAE $M^{(2)}$	bias $M^{(2)}$	cert.@1	cert.@3
Per-group (targets supervised)	24.2%	1.45	0.049	98%	0.028	0.029	-0.025	0.906	0.963
All-column (masked reconstruction)	7.7%	2.44	0.040	94%	0.024	0.026	-0.021	0.944	0.976
All-column, target held out	7.5%	2.50	0.039	94%	0.024	0.056	+0.028	0.750	0.966

$M^{(2)}$ (0.026 versus 0.029), and its certified top-1 ratio rises from 0.906 to 0.944.

The mechanism is visible in the representation. A backbone that serves one or two targets leaves a quarter of its hidden units inactive (24%) and has an effective dimension of 1.45; one that must reconstruct every column keeps 92% of units active and has an effective dimension of 2.44. It has to encode the latent operating state—load level, renewable output, congestion—that drives all grid quantities, which is exactly the argument for multi-task representation learning [34, 35, 36]. Why V improves more than M follows from (5): 89–98% of the V error is common-mode, and a better representation mostly shrinks this common part, which cancels in the marginals anyway.

The last row of Table 7 tests a target never seen in pre-training: the target column is deleted from the data before pre-training. Set values and marginals are estimated as well as with the all-column backbone, but $\hat{M}^{(2)}$ is inflated (bias +0.028), because the maximum over many marginals picks up the few that are overestimated; the certified top-1 ratio drops to 0.75. Retraining three backgrounds per field restores it to 0.966. Re-use for unseen targets therefore requires certification—which is the pipeline we use anyway.

Cost. Measured on one otherwise idle GPU, retraining the attacker family sequentially for one field set takes 14–56 s (the perceptrons dominate; one network per target), whereas the amortized estimate takes 62–78 ms, a factor of 191–763. A complete $K = 2$ table (all sets of size ≤ 3) therefore takes 2–9 min to scan, against 7–53 h of sequential retraining; certification adds three retrained backgrounds per field and target, i.e. a few hundred sets. Pre-training a backbone takes about half a minute and is done once per data set. The savings compound over the recurrent workloads of an audit: re-grading after every release, redefined targets, and the sets of size four needed for $K = 3$ disposal.

10. Results IV: Robustness and the Choice of K

Attacker seeds and data splits. Ground truth is itself an estimate: attackers are trained with random initializations and early-stopping splits, and V_y is measured on one random test split. We repeated the complete three-attacker retraining with two further seeds (RTS-GMLC, three targets; PJM load) and on two further random 70/30 splits (RTS-GMLC). Across these reruns the mean absolute change in V_y is 0.001–0.008, the Spearman correlation of $M^{(2)}$ across fields with the main run is 0.983–0.999, the set of τ -critical fields is identical at $\tau = 0.5$ (Jaccard 1.0 in every case) and agrees with Jaccard 0.93–1.00 at $\tau = 0.7$, and the number of combination-critical fields and the minimum withholding set change by at most one or two fields (Fig. 9c). The conclusions of Sections 7–8 are therefore not artefacts of a particular seed or split.

Threshold. Fig. 9(d) sweeps τ from 0.3 to 0.9 over all ten targets. At every threshold the share of fields that are safe alone but critical with at most two background fields exceeds the share of fields that are unsafe alone (e.g. 43% versus 10% at $\tau = 0.7$), and withholding the fields flagged by single-field grading leaves between 38% (at $\tau = 0.9$) and 96% (at $\tau = 0.5$) of the minimal unsafe sets intact. The gap between the two grading philosophies is thus not tied to the choice of τ .

Choice of K . Exhaustive retraining of all sets of four fields on every data set (for NEM an additional 27,405 sets) gives the exact $M^{(3)}$. Measured by the *marginal* risk, $K = 2$ is close to saturation: $M^{(2)}/M^{(3)}$, the ratio of mean field risk, lies between 0.958 and 1.000 for nine targets and is 0.813 for NEM TUMUT3, a hydro unit whose dispatch depends on inter-regional exchange (Fig. 9a). The first background field brings most of the escalation ($M^{(1)}/M^{(2)}$ between 0.73 and 0.97).

Threshold crossings, however, do not saturate as quickly, which is the caveat stated in Section 4. At $\tau = 0.7$ several targets have fields that become critical only with a third background field: PJM net interchange (3 critical fields at $K = 2$, 21 at $K = 3$), NEM PPCCGT (10 and 24) and PJM load (14 and 22) (Fig. 9b). At $\tau = 0.5$ all fields are already critical at $K = 2$ except for TUMUT3 (22 versus 30). Minimal unsafe sets of size four therefore exist that a hitting set computed at $K = 2$ does not break. Breaking all minimal unsafe sets up to size four requires on average 11.3 withheld fields instead of 9.4 at $\tau = 0.5$, and 6.3 instead of 5.1 at $\tau = 0.7$. Our recommendation

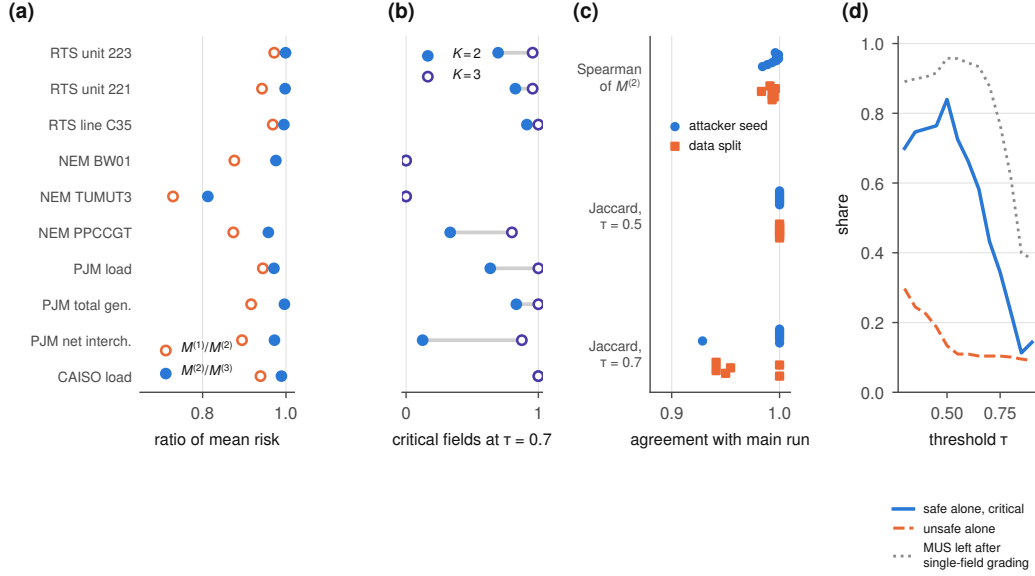


Figure 9: Robustness. (a) Saturation of mean field risk in the background budget. (b) Share of τ -critical fields at $\tau = 0.7$ with $K = 2$ (filled) and $K = 3$ (open). (c) Agreement of reruns with different attacker seeds or data splits with the main run. (d) Threshold sweep over ten targets: fields that are safe alone but critical (solid), fields unsafe alone (dashed), and minimal unsafe sets left after single-field grading (dotted).

follows: $K = 2$ suffices to *grade* fields by their marginal risk, while *disposal* decisions should be computed at $K = 3$, at the price of one to two additional withheld fields. Evaluating all sets of size four is exactly the workload for which the amortized scan is needed: for CAISO it comprises 52,360 sets per target, against 7,175 for $K = 2$.

11. Discussion

What changes for grading practice. Single-field grading is not a conservative approximation of a combination-aware audit; it fails in the unsafe direction. On every data set, most fields that take part in a dangerous combination of at most three fields are safe on their own, and withholding what single-field grading flags leaves more than 95% of those combinations intact. The field-level table proposed here keeps the familiar form of a grade per field but adds what a grade needs in order to act on it: the worst-case risk $M_{i \rightarrow y}^{(K)}$, the witness background that attains it, the escalation over single-field risk, and the

minimal unsafe sets from which a minimum withholding set follows. Because $M_{i \rightarrow y}^{(K)}$ is defined relative to the public baseline, the table is recomputed whenever something is released; the amortized attacker makes this recomputation cheap and the certification step keeps it honest.

Relation to inference-graph approaches. Propagating risk over pairwise inference relations [6] captures chains of single-field inferences and physical relations known to experts, and it does not require an attacker to be trained. It cannot represent joint inference by construction (Fig. 5), and its scores are not checkable by an attack experiment. Set-level inferability provides the missing ground truth; an inference graph could be used to propose candidate backgrounds for the scan, a combination we have not evaluated.

What the estimator is and is not. The amortized attacker is not required for correctness: every decision reported here is certified by retraining. Its role is to make the scan cheap enough to be repeated, and its accuracy matters only through the ranking of backgrounds that certification follows. The ablation shows that most of this accuracy comes from solving a readout per field set on a nonlinear conditional representation; masked pre-training chiefly improves the ordering of the largest marginals, and one pre-training on all columns serves any later choice of targets and candidates, with certification required for targets unseen in pre-training.

Limitations. (i) Risk is defined relative to a fixed attacker family; a stronger attacker can only raise V_y and hence the measured combination risk, but the numerical values are family-dependent. (ii) The adversary is assumed to hold labelled history of the target; with little auxiliary data, $M_{i \rightarrow y}^{(K)}$ as a maximum statistic is inflated by selection noise, which again argues for certified lower bounds rather than point values. (iii) We use random train/test splits; drift over time may change the values of $M_{i \rightarrow y}^{(K)}$, although not the audit procedure. (iv) $K = 2$ nearly saturates the marginal risk on nine of ten targets but not threshold crossings; disposal should use $K = 3$, and parity-like targets would defeat any fixed K . (v) Field roles for NEM, PJM and CAISO follow the Chinese disclosure rules rather than local market rules; they are used to define a realistic audit, not to assess those markets. (vi) The inference-graph baseline reimplements the propagation idea without the expert inputs and attention training of [6].

12. Conclusion

We defined field-level inference risk from set-level inferability, the out-of-sample accuracy of retrained attackers on a set of fields. The budgeted marginal risk $M_{i \rightarrow y}^{(K)}$ is the worst-case gain that releasing a field brings over any background of at most K fields, relative to what is already public; it is a threshold-free safety margin, and its critical fields are exactly the members of small minimal unsafe sets. An amortized attacker—masked pre-training with a closed-form readout per field set—scans all small sets in milliseconds, and retraining a few scanned backgrounds per field certifies the result. On four power data sets with rule-based field roles and exhaustive retraining as ground truth, single-field grading recovered 13–17% of the critical fields and left more than 95% of dangerous combinations intact, whereas the scan–certify pipeline recovered 75–88% of critical fields with no false positives and, selecting fields from estimates alone, broke all but 1–6% of dangerous combinations with close to the minimum number of withheld fields.

Data and code availability

RTS-GMLC [39] and the AEMO NEMWEB archive [40] are public; the PJM and CAISO series were obtained from the operators’ public data portals. Field-role tables, the code for dispatch simulation, ground-truth retraining, the amortized attacker and all analyses will be made available upon publication.

Appendix A. Further properties of $M_{i \rightarrow y}^{(K)}$

Proposition 3 (Ladder and candidate-set monotonicity). (a) $M_{i \rightarrow y}^{(0)} \leq M_{i \rightarrow y}^{(1)} \leq \dots \leq M_{i \rightarrow y}^{(|H|-1)}$; (b) *enlarging the candidate set from H to $H' \supseteq H$ cannot decrease $M_{i \rightarrow y}^{(K)}$* ; (c) *if f' duplicates f , i.e. $V_y(B \cup T \cup \{f'\}) = V_y(B \cup T \cup \{f\}) = V_y(B \cup T \cup \{f, f'\})$ for all T , deleting f' leaves $M_{i \rightarrow y}^{(K)}$ unchanged for every $i \neq f, f'$.*

Proof. (a) and (b) are maxima over nested feasible sets. (c) If a witness T of i contains f' , replace f' by f (if $f \notin T$) or drop it (if $f \in T$); the value is unchanged and the budget is respected, so the maximum does not decrease after deleting f' , and by (b) it does not increase. $\square$

Theorem 3 (Truncated super-efficiency). *For $A \subseteq H$ with $|A| \leq K + 1$, $V_y(B \cup A) - V_y(B) \leq \sum_{i \in A} M_{i \rightarrow y}^{(K)}$. More tightly, with $U^{(K)}(\emptyset) = 0$ and $U^{(K)}(A) = \min_{j \in A} [U^{(K)}(A \setminus \{j\}) + M_{j \rightarrow y}^{(\min(|A|-1, K))}]$, $V_y(B \cup A) - V_y(B) \leq U^{(K)}(A) \leq \sum_{i \in A} M_{i \rightarrow y}^{(K)}$.*

Proof. Order $A = \{i_1, \dots, i_m\}$ and telescope $v_B(A) = \sum_t [v_B(A_t) - v_B(A_{t-1})]$; the t -th term is a marginal of i_t over a background of size $t - 1 \leq K$, hence at most $M_{i_t \rightarrow y}^{(K)}$. The staged bound follows by induction on $|A|$ and Proposition 3(a). $\square$

For $K \geq |H| - 1$ this is the super-efficiency of MCI [7]; with a finite budget it is stronger because $M_{i \rightarrow y}^{(K)} \leq \text{MCI}$. It yields a release rule “accept A if $U^{(K)}(A) \leq \tau$ ” that never accepts an unsafe set of size $\leq K + 1$. We do not use this rule in the main text because it is conservative for larger release sets; the minimum-hitting-set disposal of Section 8 addresses the same need directly.

Proposition 4 (Irreducible synergy is bounded by field risk). *With $\text{syn}^B(S) = V_y(B \cup S) - \max_{T \subsetneq S} V_y(B \cup T)$ and $|S| \geq 2$, $\text{syn}^B(S) \leq \min_{j \in S} M_{j \rightarrow y}^{(|S|-1)}$.*

Proof. $\max_{T \subsetneq S} V_y(B \cup T) \geq V_y(B \cup S \setminus \{j\})$ for every j , hence $\text{syn}^B(S) \leq \Delta_j(B \cup S \setminus \{j\}) \leq M_{j \rightarrow y}^{(|S|-1)}$. $\square$

Remark 1 (Escalation is weaker than synergy). $M_{i \rightarrow y}^{(K)} = M_{i \rightarrow y}^{(0)}$ iff $\Delta_i(B \cup T) \leq \Delta_i(B)$ for all $|T| \leq K$. Submodularity implies no escalation but not conversely: with $V_y(\{1\}) = 0.8$, $V_y(\{2\}) = 0.6$, $V_y(\{3\}) = 0.8$, all pairs 0.8 and $V_y(\{1, 2, 3\}) = 1$, every field has $M^{(2)} = M^{(0)}$, yet $\Delta_1(\{3\}) = 0 < \Delta_1(\{2, 3\}) = 0.2$.

Appendix B. Implementation details

Attackers. Single- and multi-target perceptrons: two hidden layers of 128 ReLU units, dropout 0.15, Adam (learning rate 10^{-3} , weight decay 5×10^{-4}), batch 128, at most 300 epochs, patience 60, epoch chosen per (set, target) on validation. Many attackers are trained in parallel on one GPU as independent networks whose inputs are masked to their own field set; masked columns receive no gradient, so each network is functionally a network trained on its set alone. Gradient boosting: histogram gradient boosting with (learning rate,

leaves) $\in \{(0.05, 15), (0.1, 31)\}$, 300 iterations with internal early stopping; the configuration is chosen on validation. The family member is chosen per (set, target) on validation. Across data sets gradient boosting is selected for 35–76% of (set, target) pairs (RTS-GMLC 76%, NEM 35%, PJM 67–69%, CAISO 52%), so the family maximum matters.

Estimator. Backbone: three hidden layers of 256 ReLU units with dropout 0.1, input $[x \odot m, m]$, Adam (10^{-3} , weight decay 5×10^{-4}), batch 256, at most 400 epochs, patience 60 on eight fixed validation masks. Ridge penalties $\lambda \in \{10^{-3}, \dots, 10^2\}$.

Inference-graph baseline. Following the structure of [6] without its expert-assigned intrinsic scores: nodes are the candidate fields and the target; the weight of edge $i \rightarrow j$ is the validation R^2 of predicting field j from field i alone by gradient boosting; the score of i is $\sum_{h=1}^3 0.5^{h-1} (W^h)_{i,y}$. The first term equals single-field predictability; longer paths add inference chains through other fields.

SAGE. Marginal imputation with 32 background rows, 512 evaluation rows from the validation split and 24 random permutations, on a gradient-boosting model fitted to all candidates.

Table B.8: Field roles and rule basis (Chinese electricity-market disclosure rules [1]; appendix item numbers in parentheses).

Data set	Candidate fields (released before the target)	Targets
RTS-GMLC	zonal load forecasts (3; 6.41); wind, PV, rooftop PV and hydro forecasts (4; 6.42); regulation up/down and spinning reserve requirements (3; 6.46); system energy price and nodal prices at 9 buses (10; 6.52); congestion indicators of lines C6, C29, CA-1 (3; 6.54)	outputs of units 223_STEAM_3 and 221_CC_1 (specific information, Art. 17(3)); hourly flow on line C35 (only daily averages of important lines are listed, 6.66)
NEM 2024	regional demand (5), available capacity (5), semi-scheduled wind and solar forecasts (9), regional prices (5), interconnector flows (6)	actual outputs of units BW01, TUMUT3, PPCCGT (specific information)
PJM	regulation market prices and volumes (5); day-ahead reserve prices, requirements, self-schedules and cleared volumes (9); day-ahead system energy price, LMP, congestion and loss components (4); real-time LMP, congestion and loss components (3); scheduled interchange (1); for total generation and net interchange also the system load forecasts (2)	metered load (6.61), total generation (6.57), net actual interchange (6.65), all released next day
CAISO	day-ahead load forecasts of three sub-areas (3); day-ahead wind and solar forecasts (6); day-ahead and 15-min real-time prices and components (11); day-ahead schedules of sub-area load, total generation, imports and exports (6); ancillary-service requirements incl. regulation mileage (5) and procured volumes (4)	actual CAISO load (6.61), released next day

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